

Mohammad Reza Babaei Mosleh | Curriculum Vitae

Institut Polytechnique de Paris—Department of Electrical Engineering

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Qualification Profile

Master's student at Institut Polytechnique de Paris with 3+ years of professional experience spanning AI engineering, full-stack software development, and signal and information processing across multiple industry sectors. Expert Python developer with production experience building LLM-powered platforms, agentic AI workflows, RAG architectures, and scalable distributed backend systems deployed on GCP and AWS, complemented by full-stack development experience in React and Next.js. Proven leader who has managed teams of engineers, driven system design decisions, and delivered measurable performance improvements through full-stack engineering, data pipelines, and database management. Published researcher in applied machine learning with work spanning medical signal processing, IoT malware classification, and Parkinson's disease detection. Combines strong software engineering fundamentals with genuine curiosity for AI, cloud systems, and emerging technologies, and is comfortable operating autonomously in cross-functional, fast-moving teams.

SKILLS

- **Programming & Scripting:** Python (Expert), JavaScript/TypeScript (Node.js, Next.js, React), C/C++, Bash
- **AI & ML:** LlamaIndex, LangChain(LangGraph), Prompt Engineering, Prompt Optimization
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis
- **Architecture & Infrastructure:** System Design, Cloud Systems, Distributed Systems, Linux, Kafka, RabbitMQ
- **Cloud & DevOps Tools:** Google Cloud Platform(GCP)/Azure Web Services (AWS), Git/GitHub, Postman, AI LLM Tools
- **Project Management Tools:** Jira, Trello
- **Collaboration & Documentation Tools:** Confluence, Notion, LaTeX
- **Engineering Tools:** Tableau, STM32CubeIDE, Zephyr, Vivado, Xilinx ISE Designer, MATLAB, Microwind
- **Scientific Foundations:** Signal Processing, Time-Series Analysis, Biomedical AI, Information Theory, Statistical Learning

WORK EXPERIENCE

- **SAMOVAR LAB** – Research Intern July 2026 – September 2026
 - Conducted research on human gait analysis using IMU sensor data to distinguish individuals with Parkinson's disease from healthy control subjects.
 - Developed deep learning models based on Bi-LSTM autoencoders and temporal attention mechanisms, achieving the highest reported classification accuracy on the evaluated dataset.
 - Performed cross-task evaluation to assess model generalization across different gait activities and temporal analysis to investigate behavioral patterns and progression differences between Parkinson's and control groups.
 - Contributed to the experimental design, analysis, and manuscript preparation of a research paper based on the developed methodology and findings.
- **PropertyGPT** – Software Developer and AI Engineer June 2023 – June 2026
 - Architected and deployed an end-to-end AI-powered real estate platform serving personalized home-search and investment tours, integrating LLM-driven insight extraction via LangGraph and LlamaIndex with Agentic AI workflows and solid backend infrastructure to help AI agents and make the system more efficient.
 - Deployed backend services in both Node.js and Python, serving as different processing units for the Agentic workflow.
 - Built and maintained scalable data pipelines to scrape, clean, and store over 3 million real estate listings using Python, GCP Cloud Run and Storage, PostgreSQL and MongoDB.
 - Developed RESTful backend services with FastAPI, containerized with Docker and orchestrated via Kubernetes, deployed on GCP/AWS.
 - Optimized LLM query pipelines and retrieval-augmented generation (RAG) architecture, achieving approximately 2x improvement in response speed.
 - Optimized data population pipeline, achieving more than 10x speed efficiency (considering about 100GB of

- data).
- Collaborated within cross-functional teams of 20+ engineers and product stakeholders, contributing to system design decisions and agile sprint planning.
- Led a team of 7 engineers, overseeing task allocation, code reviews, and technical delivery.
- o **NikTed** – Embedded System Engineer and AI Engineer July 2023 – September 2023
 - Optimizing frequency-based signal amplification for home-care medical devices in the R&D department, with a focus on hearing aid hardware.
 - Developed a computer vision pipeline to automatically detect and count white blood cells from microscopy images, enabling early-stage disease screening.
 - Programmed embedded firmware using STM32CubeIDE and Zephyr RTOS for low-power medical devices.
- o **IthermAI** – AI Engineer and Database Administrator July 2022 – February 2023
 - Managed and curated large-scale image datasets for an AI-based smoke and fire detection system used in industrial camera deployments.
 - Designed and maintained database schemas and data ingestion workflows to support model training and evaluation pipelines.
 - Collaborated with the ML team to label, validate, and version datasets, ensuring data quality and consistency across training cycles.

EDUCATION

- o **M2 Master of Science** 2026–Present
Institut Polytechnique de Paris Paris, France
 - Master in Information Processing – Machine Learning, Communications, and Security
- o **M1 Master of Science** 2025–2026
Institut Polytechnique de Paris Paris, France
 - Electrical Engineering for Communications and Information Processing
 - Major: DATA Analysis and Pattern Classification (Datapac) – GPA: 17.4/20 (40 credits)
- o **Bachelor of Science** 2019–2024
Amirkabir University of Technology Tehran, Iran
 - Electrical and Computer Engineering
 - Major: Electronics, Minor: Computer (Artificial Intelligence) – GPA: 3.83/4 (18.20/20, 140 credits)

PUBLICATIONS

- o **M.R Babaei Mosleh**, S. Sharifian (2024). An Efficient Cloud-Integrated Multi-Stage Convolutional Neural Network Framework Approach to IoT Malware Classification (**published**).
- o **M.R. Babaei Mosleh**, N. Houmani, S. Garcia-Salicetti (2026). Human Gait Analysis for Parkinson Disease Classification Utilizing Bi-LSTM and Temporal Attention-Based Neural Networks on IMU Sensor Data (**under submission**).

LANGUAGE SKILLS

- o English: Fluent (C1)
- o French: Basic (A2)